


Problem sheet. 1.

$$\begin{aligned}
 1. \quad E &= \int_{\mathbb{R}} \frac{1}{4\pi\epsilon_0} \frac{r-r'}{|r-r'|^3} \cdot \lambda(x') dx' \\
 &= \frac{1}{4\pi\epsilon_0} \int_{\mathbb{R}} \frac{(-x', 0, r)}{(x'^2 + r^2)^{3/2}} \cdot \lambda(x') dx' \\
 &= \frac{1}{2\pi\epsilon_0} \int_0^L \frac{(x, 0, r)}{(x^2 + r^2)^{3/2}} \alpha \cdot x^2 dx \\
 &= \frac{\frac{1}{2}}{2\pi\epsilon_0} \int_0^L \frac{\alpha r \cdot x^2}{(x^2 + r^2)^{3/2}} dx
 \end{aligned}$$

$$\begin{aligned}
 x &= r \tan \theta \\
 &= \frac{\frac{1}{2}}{2\pi\epsilon_0} \int_0^{\arctan \frac{L}{r}} \frac{\alpha \cdot r \cdot r^3 \tan^3 \theta}{r^3} \cdot \cos^3 \theta \cdot r \cdot \frac{1}{\cos^2 \theta} d\theta \\
 &= \frac{\alpha r^2}{2\pi\epsilon_0} \int_0^{\arctan \frac{L}{r}} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta \\
 &= -\frac{\alpha r^2}{2\pi\epsilon_0} \int_0^{\arctan \frac{L}{r}} \frac{1 - \cos^2 \theta}{\cos^2 \theta} d \cos \theta \\
 &= \frac{\alpha r^2}{2\pi\epsilon_0} \left[\frac{1}{\cos \theta} + \cos \theta \right]_0^{\arctan \frac{L}{r}}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad (a) \quad E_n &= \int_0^\pi \frac{1}{4\pi\epsilon_0} \frac{r-r'}{|r-r'|^3} \cdot \lambda \cdot R \cdot d\theta \\
 &= \frac{\lambda R}{4\pi\epsilon_0} \int_0^\pi \frac{(R \cos \theta, R \sin \theta, r)}{(\sqrt{R^2 + r^2})^3} d\theta \\
 &= \frac{\lambda R}{4\pi\epsilon_0} \cdot \frac{1}{(R^2 + r^2)^{3/2}} \left(\hat{y} \cdot \int_0^\pi -R \sin \theta d\theta + \frac{1}{2} \cdot \int_0^\pi r d\theta \right) \\
 &= \frac{\lambda R}{4\pi\epsilon_0} \cdot \frac{1}{(R^2 + r^2)^{3/2}} \left(-2R \cdot \hat{y} + \pi r \frac{1}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } E_b &= 2 \cdot \frac{\lambda R}{4\pi\epsilon_0} \cdot \frac{1}{(R^2+r^2)^{3/2}} \cdot \pi r \cdot \frac{1}{2} \\
 &= \frac{\lambda}{2\epsilon_0} \frac{Rr}{(R^2+r^2)^{3/2}} \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } E_c &= \frac{1}{2} \int_0^R \frac{1}{2\epsilon_0} \cdot \frac{xr}{(x^2+r^2)^{3/2}} \lambda \cdot dx \\
 &= \frac{1}{2} \int_0^R \frac{1}{2\epsilon_0} \cdot \frac{xr}{(x^2+r^2)^{3/2}} \sigma \cdot dx \\
 &= \frac{\sigma r}{2\epsilon_0} \int_0^R \frac{1}{\sqrt{x^2+r^2}} \cdot dx \\
 &= \frac{\sigma r}{2\epsilon_0} \left(\frac{1}{\sqrt{R^2+r^2}} - \frac{1}{r} \right) \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \lambda &= \frac{[\pi(r+dr)^2 - \pi r^2] \cdot \sigma}{2\pi r} \\
 &= \frac{1}{2\pi r} \sigma \pi \cdot 2r \cdot dr = \sigma \cdot dr
 \end{aligned}$$

$$3. \quad \text{a) } E = \frac{q}{4\pi\epsilon_0} \frac{r-r_0}{|r-r_0|^3}$$

$$\begin{aligned}
 \nabla \cdot E &= q \frac{1}{4\pi\epsilon_0} \cdot 4\pi \delta(r-r_0) \\
 &= \frac{q}{\epsilon_0} \delta(r-r_0)
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \rho(r) = \sum q_i \delta(r-r_i) &\Leftrightarrow Q(V) = \iiint_V \rho \, dV = \iiint_V q_i \delta(r-r_i) \, dV \\
 &= q_{\text{end}}
 \end{aligned}$$

$$\text{c) } \nabla \cdot E = \frac{\rho}{\epsilon_0}$$

$$\iint_{\partial V} E \cdot d\vec{s} = \iiint_V (\nabla \cdot E) \, dV = \iiint_V \frac{\rho}{\epsilon_0} \, dV = \frac{1}{\epsilon_0} \cdot q_{\text{end}}$$

for a R-ball

$$E(r) = \frac{1}{4\pi\epsilon_0 r^2} Q_{\text{end}} = \begin{cases} \frac{\rho R}{3\epsilon_0} \cdot \hat{r} & |\hat{r}| < R \\ \frac{\rho R}{\epsilon_0} \cdot \hat{r} & |\hat{r}| \geq R \end{cases}$$

cancel the ball.

$$E'(r) = \begin{cases} 0 \\ \frac{\rho}{3\epsilon_0} \frac{r^3 - R_1^3}{r^3} r \\ \frac{\rho}{3\epsilon_0} \frac{R_2^3 - R_1^3}{r^3} r \end{cases}$$

Problem Sheet 2.

$$\begin{aligned}
 1. \quad \iint_S \vec{E} \cdot \vec{n} \, ds &= \int_{-1}^1 \int_{-1}^1 (y^2 + z^2, x^2 + z^2, x^2 + y^2) \cdot (0, 0, 1) \, dx \, dy \\
 &= \int_{-1}^1 \int_{-1}^1 x^2 + y^2 \, dx \, dy \\
 &= \int_{-1}^1 2y^2 + \frac{2}{3} \, dy \\
 &= \frac{4}{3} + \frac{4}{3} = \frac{8}{3}
 \end{aligned}$$

$$2. \quad \nabla \cdot \vec{E} = \nabla \cdot \left[\left(\frac{E(r)}{r} \right) \vec{r} \right] = 3 \frac{E(r)}{r} + r \frac{d\left(\frac{E(r)}{r}\right)}{dr} = \frac{2E}{r} + \frac{dE}{dr}$$

$$\begin{aligned}
 3. \quad (a) \quad \oint_{\partial V} \vec{E} \cdot \vec{n} \, ds &= \frac{1}{\epsilon_0} \iiint_V \rho \, dV \\
 4\pi R^2 \cdot E &= \frac{1}{\epsilon_0} \iiint_V \alpha r \cdot e^{-\beta r^2} \, dV \\
 &= \frac{\alpha}{\epsilon_0} \int_0^R \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} r^3 \cdot e^{-\beta r^2} \cos \theta \, d\theta \, d\phi \, dr \\
 &= \frac{4\pi\alpha}{\epsilon_0} \int_0^R r^3 e^{-\beta r^2} \, dr \\
 &= \frac{4\pi\alpha}{\epsilon_0} \int_0^{R^2} \frac{1}{2} t \cdot e^{-\beta t} \, dt \\
 &= \frac{2\pi\alpha}{\epsilon_0} \left[-\frac{1}{\beta} t \cdot e^{-\beta t} - \frac{1}{\beta^2} e^{-\beta t} \right]_0^{R^2} \\
 E &= \frac{\alpha}{2\epsilon_0\beta^2} \left[\frac{1}{r^2} (1 - e^{-\beta r^2}) - \beta e^{-\beta r^2} \right] \hat{r}
 \end{aligned}$$

$$1b) \quad \Phi = \frac{1}{\epsilon_0}$$

$$\begin{aligned} (c) \quad \nabla \cdot \vec{E} &= \frac{1}{2\epsilon_0 \cdot \tilde{r}^2} \cdot \left(2 \frac{E}{r} + \frac{dE}{dr} \right) \\ &= \frac{\rho}{\epsilon_0} \end{aligned}$$

Problem Sheet 3.

$$1. \oint_{\partial V} \vec{E} \cdot \vec{n} \, ds = \frac{q}{\epsilon}$$

$$q = -E \cdot 4\pi R^2 \cdot \epsilon_0$$

$$2. (a) \nabla \times \left(E \frac{\vec{r}}{r} \right) = \nabla \left(\frac{E}{r} \right) \times \vec{r} + \frac{E}{r} \nabla \times \vec{r}$$

$$= \frac{rE'(r) - E(r)}{r^2} \left(\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z} \right) \times (x, y, z)$$

$$= 0$$

$$1b) \nabla \times \vec{E} = \nabla (E) \times \vec{r} + E \cdot \nabla \times \vec{r}$$

$$= E'(r) \cdot \left(\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z} \right) \times \vec{r}$$

$$= \vec{0}$$

$$\nabla \times (\nabla F) = 0$$

$$\nabla \cdot (\nabla \times F) = 0$$

$$\nabla V = -\vec{E}$$

$$V = \frac{\alpha e^{-\beta r^2}}{2\beta}$$

$$(c) \nabla \cdot (\vec{E}) = \nabla (E \cdot \vec{r}) = 3E + r \frac{dE}{dr}$$

$$= 3\alpha e^{-\beta r} + r(2\beta) e^{-\beta r}$$

$$\rho(r) = \alpha \epsilon_0 (3 - 2\beta r^2) e^{-\beta r^2}$$

$$3. (a) \nabla \cdot E = \frac{\rho}{\epsilon_0}$$

$$2. \frac{E}{r} + \frac{dE}{dr} = \frac{\rho}{\epsilon_0}$$

$$\frac{d}{dr} (Er) = \frac{r\rho}{\epsilon_0}$$

$$E = \begin{cases} \frac{\alpha}{5\epsilon_0} r^3 \hat{r} & r < R \\ \frac{\alpha}{5\epsilon_0} \frac{R^4}{r^2} \hat{r} & r \geq R \end{cases}$$

$$b) \quad q = \iiint p(r) \, dr = \begin{cases} \frac{4\pi a}{r} k^5 \\ \frac{4\pi a}{r} r^5 \end{cases}$$

Problem Sheet 4.

$$\begin{aligned}
 1. (a) \quad V(r) &= \frac{1}{4\pi\epsilon_0} \left(\int_{B(r+dr)} - \int_{B(r)} \right) \left(\frac{\rho}{|r-r'|} dV \right) \\
 &= \frac{1}{4\pi\epsilon_0} \left(\int_{B(r+dr)} - \int_{B(r)} \right) (\rho dV) \frac{1}{|r|} = \begin{cases} \frac{\sigma R^2}{\epsilon_0 |r|} \cdot \hat{r} \\ \frac{\sigma R^2}{\epsilon_0 |r|} \cdot \hat{r} \end{cases}
 \end{aligned}$$

(b) Dinit

$$2. (a) \quad \oint_{\partial V} \mathbf{D} \cdot d\mathbf{a} = L\lambda$$

$$\mathbf{D} = \frac{\lambda}{2\pi r} \hat{r}$$

$$\mathbf{E} = \begin{cases} \frac{\lambda}{2\pi\epsilon_0 (1+\chi_e)r} \hat{r} & |r| < R \\ \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} & |r| > R \end{cases}$$

$$(b) \quad ① \quad \epsilon_0 \cdot \nabla \cdot \mathbf{E} = \rho_b + \rho_f$$

$$② \quad \rho_b = -\nabla \cdot \mathbf{P} = -\epsilon_0 \chi_e \nabla \cdot \mathbf{E}$$

$$\chi_e ① + ② : (1+\chi_e) \rho_b = -\chi_e \rho_f$$

$$\rho_b = -\frac{\chi_e}{1+\chi_e} \lambda \delta^w(r)$$

$$\int_V \rho_b dV + \int_{\partial V} \sigma_b dA = 0$$

$$\int_V \rho_b dV = -\frac{\chi_e \lambda}{1+\chi_e} \int_0^L \int_{\mathbb{R}^2} \delta^w(r) dr = -\frac{\chi_e}{1+\chi_e} \lambda L$$

$$\sigma_b = \frac{\chi_e}{1 + \chi_e} \cdot \frac{\lambda}{2\pi R}$$

$$3. \text{I} \quad Q_v(t) = \int_V \rho(\vec{r}, t) dV$$

$$\begin{aligned} \frac{dQ_v(t)}{dt} &= \frac{\partial}{\partial t} \int_V \rho(\vec{r}, t) dV = \int_V \frac{\partial \rho}{\partial t} dV \\ &= \oint_{\partial V} \vec{J} \cdot \vec{n} da \quad \text{pointing outside.} \\ &= \int_V \nabla \cdot \vec{J} dV \end{aligned}$$

$$\text{thus: } \frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{J}$$

$$\text{II. } \iint_{\mathbb{R}^3 \times \mathbb{R}} \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} \right) \varphi d\vec{r} dt = 0 \quad \text{in distribution meaning.}$$

$$\iint \frac{\partial \rho}{\partial t} \varphi d\vec{r} dt = - \sum \int q_i \partial_i \varphi(r_i(t), t) dt$$

$$\left[\frac{d}{dt} \varphi(r_i(t), t) = \partial_t \varphi + \dot{r}_i \cdot \nabla \varphi \right]$$

$$= - \sum_i q_i \int \frac{d}{dt} \varphi(r_i(t), t) dt + \sum_i q_i \int \dot{r}_i \cdot \nabla \varphi(r_i(t), t) dt$$

$$\left[\begin{aligned} \varphi \text{ testing function} &\Rightarrow \varphi(\cdot, t) \rightarrow 0 \quad (t \rightarrow \pm \infty) \\ &\Rightarrow \int_{-\infty}^{\infty} \frac{d}{dt} \varphi(\cdot, t) dt = 0 \end{aligned} \right]$$

$$\text{thus } \int (\partial_t \rho) \varphi = \sum_i q_i \int \dot{r}_i \cdot \nabla \varphi(r_i(t), t) dt$$

$$\begin{aligned}
 \int (\nabla \cdot \mathbf{J}) \varphi &= - \int \mathbf{J} \cdot \nabla \varphi \\
 &= - \sum_i \iint q_i \dot{\mathbf{r}}_i(t) \delta(\mathbf{r} - \mathbf{r}_i) d\mathbf{r} dt \\
 &= - \sum_i q_i \int \dot{\mathbf{r}}_i \cdot \nabla \varphi(\mathbf{r}_i(t), t) dt.
 \end{aligned}$$

thus $\iint (\partial_t \rho + \nabla \cdot \mathbf{J}) \varphi = 0$
 which also implies $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$

back to the question.

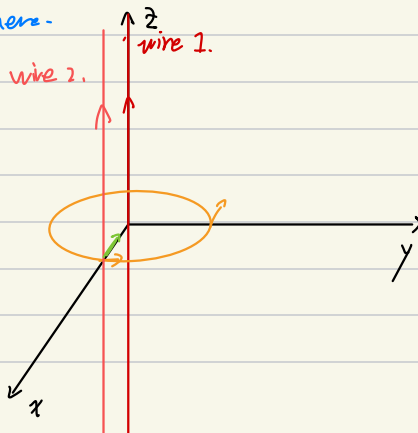
$$\frac{\partial}{\partial t} \rho = - \nabla \cdot \mathbf{J} = - \frac{\alpha \ell}{\epsilon} e^{-\beta t} \cdot f(r)$$

$$f(r) = \nabla \cdot (r^2 \cdot \vec{r}) = 3r^2 + r \frac{dr^2}{dr} = 4r^2 \quad (r < R)$$

$$f(r) = \nabla \cdot \left(\frac{\vec{r}}{r^3} \right) = 2\pi \delta^{(2)}(\vec{r}) \quad (r > R)$$

$$\rho(r, t) = \begin{cases} \alpha r^2 (1 + e^{-\beta t}) & (r < R) \\ 0 & (r > R) \end{cases}$$

Magnetic Starts here.



$$B(r) = \frac{\mu_0}{4\pi} \int_C \frac{I dr' \times (r-r')}{|r-r'|^3}$$

$$\int J \cdot d\tau = I \int dr'$$

$$[J = \sum_i q_i \vec{v}_i \delta(r-r_i)]$$

$$\text{then } B(r) = \frac{\mu_0}{4\pi} \int_V \frac{J(r') \times (r-r')}{|r-r'|^3} d\tau'$$

$$= \frac{\mu_0}{4\pi} \nabla \times \int_V \frac{J(r')}{|r-r'|} d\tau'$$

$$= \nabla \times A$$

$$[A = \frac{\mu_0}{4\pi} \int \frac{J(r')}{|r-r'|} d\tau']$$

$$\nabla \times B = \frac{\mu_0}{4\pi} (\nabla (\nabla \cdot A) - \nabla^2 A)$$

$$= \frac{\mu_0}{4\pi} [\underbrace{\nabla \cdot \int \frac{J(r')}{|r-r'|} d\tau'}_{I_1} - \underbrace{\nabla^2 \int \frac{J(r')}{|r-r'|} d\tau'}_{I_2}]$$

$$I_1 = \nabla \cdot \left[\underbrace{(\nabla \cdot J)}_0 \frac{1}{|r-r'|} + J \cdot \nabla \frac{1}{|r-r'|} d\tau' \right]$$

$$= \nabla \cdot \left(\int J \cdot \nabla \frac{1}{|r-r'|} d\tau' \right) [\text{where } \nabla \frac{1}{|r-r'|} = -\nabla' \frac{1}{|r-r'|}]$$

$$= \nabla \cdot \left[\int \nabla' \cdot \left(\frac{J}{|r-r'|} \right) - \underbrace{(\nabla' \cdot J)}_0 \frac{1}{|r-r'|} d\tau' \right]$$

0 - assume constant current.

$$= -\nabla \cdot \int \nabla' \left(\frac{J}{|r-r'|} \right) d\tau' = -\nabla \cdot L$$

$$L = \lim_{R \rightarrow +\infty} \int_{B_R(0)} \nabla' \left(\frac{J}{|r-r'|} \right) d\tau' = \lim_{R \rightarrow +\infty} \oint_{\partial B_R(0)} \frac{J}{|r-r'|} \cdot \vec{n} da$$

$$= 0$$

since Physicists retain $A \rightarrow 0$ or $A \sim \frac{1}{r}$ as $r \rightarrow +\infty$

$$I_x = - \left(\nabla \cdot \int \nabla \left(\frac{J_x}{|r-r'|} \right) d\tau' \right)_x$$

$$= - \nabla \cdot \int J_x(r') \cdot \frac{r-r'}{|r-r'|^3} d\tau'$$

$$= - \int J_x(r') \cdot 4\pi \delta^{(3)}(r-r') d\tau'$$

$$= 4\pi J$$

$$\text{thus } \nabla \times B = \mu_0 J$$

$$A(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{J(r')}{|r-r'|} d\tau' = \frac{\mu_0 I}{4\pi} \int_{\partial S} \frac{1}{|r-r'|} d\vec{r}'$$

$$\approx \frac{\mu_0 I}{4\pi r} \oint_{\partial S} (\hat{r} \cdot \vec{r}') d\tau' = I' \cdot \frac{\mu_0 I}{4\pi r}$$

$$\left[\oint_{\partial S} F d\vec{r}' = - \int_S \begin{vmatrix} i & j & k \\ \partial_x F & \partial_y F & \partial_z F \\ dy dz & dz dx & dx dy \end{vmatrix} \right]$$

$$= - \int_S (\nabla F \times \vec{n}) dS$$

$$I' = \oint_{\partial S} (\hat{r} \cdot \vec{r}') d\tau' = - \int_S \nabla'(\hat{r} \cdot \vec{r}') \times \vec{n} dS' = - \int_S \hat{r} \times \vec{n} dS$$

$$= - \hat{r} \int dS$$

$$A(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

$$\text{where } \vec{m} = I \int_S d\vec{a}'$$

$$\nabla \times F = G$$

$$\nabla \times (F + \nabla A) = G$$

$$\nabla \cdot F = 0$$

$$\nabla \cdot F + \nabla^2 A = 0$$

$$\nabla \times (\nabla \times F) = \nabla (\nabla \cdot F) - \nabla^2 F = \nabla \times G$$

$$\nabla^2 F = -\nabla \times G$$

$$F = \frac{1}{4\pi} \int \frac{\nabla' \times G}{|\mathbf{x} - \mathbf{x}'|} d\tau'$$

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} \nabla' \times G(\mathbf{x}') + \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) \times G(\mathbf{x}') = \nabla' \times \left(\frac{G(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right)$$

$$\left[\iiint_V \nabla \times F d\tau = - \iint_{\partial V} F \times \vec{n} dS \right]$$

$$\iiint_{\mathbb{R}^3} \nabla' \times \left(\frac{G(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right) d\tau = \lim_{R \rightarrow +\infty} \int_{B_R(0)} \nabla \times \left(\frac{G(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right) d\tau$$

$$= \lim_{R \rightarrow +\infty} - \oint_{\partial B_R(0)} \frac{G(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \times \vec{n} dS$$

$$\stackrel{(*)}{\sim} - \lim_{R \rightarrow +\infty} \hat{e} \left| \frac{G(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \times \frac{\mathbf{x}'}{R} \right| \cdot 4\pi R^2$$

$$G(\mathbf{x}') \leq \frac{C}{|\mathbf{x}'|^{1+\epsilon}}$$

$$\stackrel{(**)}{\sim} - \lim_{R \rightarrow +\infty} \hat{e} \frac{1}{R} \cdot \frac{1}{R^{2+\epsilon}} \cdot 4\pi R^2 = \vec{0}$$

thus
$$\int \frac{\nabla' \times G}{|x-x'|} = - \int \nabla' \left(\frac{1}{|x-x'|} \right) \times G$$

$$F(x) = \frac{1}{4\pi} \int G(x') \times \frac{x-x'}{|x-x'|^3} d\tau'$$

$$B = \frac{\mu_0}{4\pi} \int \frac{J(r') \times (r-r')}{|r-r'|^3} d\tau' = \nabla \times A$$

$$\nabla \times B = \mu_0 \cdot J$$

$$B = \nabla \times A \quad A = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2} = \frac{\mu_0 I}{4\pi} \oint \frac{1}{|r-r'|} dr'$$

Problem Sheet 5.

2.
$$B(r) = \frac{\mu_0 I}{4\pi} \int_C \frac{dr' \times (r-r')}{|r-r'|^3} \quad r' = (R \cos t, R \sin t, 0)$$

$$= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{(-R \sin t, R \cos t, 0) \times (r_x - R \cos t, r_y - R \sin t, r_z)}{|r-r'|^3} dt$$

$$= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{1}{|r-r'|^3} \begin{vmatrix} i & j & k \\ -R \sin t & R \cos t & 0 \\ -R \cos t & -R \sin t & r_z \end{vmatrix} dt$$

$$= \frac{\mu_0 I}{4\pi} \int_0^{2\pi} \frac{(R r_z \cos t, R r_z \sin t, R^2)}{(r_z^2 + R^2)^{3/2}} dt$$

$$= \frac{\mu_0 I}{2} \cdot \frac{R^2}{(R^2 + r_z^2)^{3/2}} \cdot \frac{1}{2}.$$

2.

$$\begin{aligned}
 B(r) &= \frac{\mu_0 I}{4\pi} \int_C \frac{dr' \times (r-r')}{|r-r'|^3} \\
 &= \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{(0,0,1) \times (r_x, r_y, r_z-t)}{|(r_x, r_y, r_z-t)|^3} dt \\
 &= \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{(-r_y, r_x, 0)}{(t^2 + r^2 - 2tr_z)^{3/2}} dt \\
 &= \frac{\mu_0 I}{4\pi} (-r_y, r_x, 0) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\frac{\sqrt{r_x^2 + r_y^2}}{((r_x^2 + r_y^2) \frac{1}{\cos^3 \theta})^{3/2}} \cdot \frac{1}{\cos \theta} d\theta}{t = \sqrt{r_x^2 + r_y^2} \tan \theta + r_z} \\
 &= \frac{\mu_0 I}{2\pi} \cdot \frac{1}{r_x^2 + r_y^2} (-r_y, r_x, 0)
 \end{aligned}$$


$$\begin{aligned}
 \nabla \times A &= \frac{\mu_0 I}{2\pi} \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, 0 \right) \\
 &= (\partial_y A_z - \partial_z A_y, \partial_z A_x - \partial_x A_z, \partial_x A_y - \partial_y A_x) \\
 &= (-\partial_z A_y, \partial_z A_x, \partial_x A_y - \partial_y A_x)
 \end{aligned}$$

$$A_y = \frac{yz}{x^2 + y^2} \cdot C \quad A_x = \frac{xz}{x^2 + y^2} \cdot C \quad A_z = 0$$

$$\begin{aligned}
 A &= \frac{1}{4\pi} \int \frac{B(r') \times (r-r')}{|r-r'|^3} d\tau' \\
 &= \frac{1}{4\pi} \int \frac{\nabla' \times B(r')}{|r-r'|} d\tau' \\
 &= \frac{\mu_0 I}{2\pi} \ln \frac{R}{r} \hat{z}
 \end{aligned}$$

3. (a) $B(\vec{r}) = \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} \frac{J(r') \times (r-r')}{|r-r'|^3} d\tau'$

$$B_z(\vec{r}) = \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} 0 \, d\tau' = 0$$

$$\begin{aligned}
 (b) \quad & \int_{z_1}^{z_2} \int_0^{2\pi} \mathbf{B} \cdot \hat{\mathbf{r}} \, R' \, d\theta \, dR = 2\pi R' (z_2 - z_1) B_r \\
 & = \int_V \nabla \cdot \mathbf{B} \, d\tau = 0
 \end{aligned}$$

$$(c) \quad \oint \mathbf{B} \cdot d\hat{\mathbf{r}} = 2\pi r B = \mu_0 J = \frac{\mu_0 J \hat{\theta}}{2} \left\{ \begin{array}{l} r \\ \frac{R^2}{r} \end{array} \right.$$

Problem Sheet 6.

$$1. \quad m \cdot \hat{r} = \frac{mz}{r}$$

$$\mathbf{B} = \nabla \times \mathbf{A} = \left(-\frac{\partial}{\partial z} A_y, \frac{\partial}{\partial z} A_x, \frac{\partial}{\partial x} A_y - \frac{\partial}{\partial y} A_x \right)$$

$(A_z = 0)$

$$A_x = -\frac{m\mu_0 y}{4\pi r^3} + G_1(x, y)$$

$$A_y = -\frac{m\mu_0 x}{4\pi r^3} + G_2(x, y)$$

$$\mathbf{A} = \frac{\mu_0}{4\pi} \frac{m \times \hat{r}}{r^2}$$

$$2. \quad \nabla \times \mathbf{E} = 0 \quad (\text{electrostatics})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{electrodynamics})$$

$$\nabla \times (\mathbf{E} + \mathbf{X}) = 0$$

$$\nabla \times \mathbf{X} = \frac{\partial \mathbf{B}}{\partial t} = \frac{\partial}{\partial t} (\nabla \times \mathbf{A}) = \nabla \times \frac{\partial \mathbf{A}}{\partial t}$$

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0$$

$$3. \quad \vec{E} = E_0 \sin(k(z - ct)) \hat{z} \quad \text{is a wave equation.}$$

$$\nabla \cdot \mathbf{E} = E_0 k \cos(k(z - ct)) \neq 0$$

$$4. (a) \quad \frac{\partial^i E}{\partial t^i} = \vec{E}_0 \omega^i f''(\omega t - \vec{k} \cdot \vec{r})$$

$$\nabla^i E = \vec{E}_0 k^i f''(\omega t - \vec{k} \cdot \vec{r})$$

$$\omega^2 = c^2 k^2$$

$$(b) \quad \nabla \cdot \vec{E} = 0$$

$$\vec{E}_0 \cdot \vec{k} = 0$$

Problem Sheet 7.

$$1. \nabla \times \mathbf{E}_1 = \alpha \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ xyz & (x-y)z & (x-y)y \end{vmatrix} = \alpha \underset{\neq \vec{0}}{(x-2y-x+2y, x-y, z-0)}$$

$$\nabla \times \mathbf{E}_2 = \alpha \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ yz & (x-2y)z & (x-y)y \end{vmatrix} = \alpha (0, 0, 0) = \vec{0}$$

$$\mathbf{E} = -\nabla V$$

$$\begin{aligned} V &= -\alpha [xyz + V(y, z)] \\ &= -\alpha [xyz - y^2 z + V(z)] \\ &= -\alpha (xyz - y^2 z) \end{aligned}$$

$$\begin{aligned} 2. (a) \nabla \cdot \mathbf{E} &= \frac{\rho(r)}{\epsilon_0} \\ &= 3\alpha r e^{-\beta r} + r [\alpha e^{-\beta r} + \alpha r (-\beta e^{-\beta r})] \\ &= 4\alpha r e^{-\beta r} - \alpha \beta r^2 e^{-\beta r} \end{aligned}$$

$$\begin{aligned} q_{\text{enc}} &= \int_V \epsilon_0 (4\alpha r e^{-\beta r} - \alpha \beta r^2 e^{-\beta r}) dV \\ &= \epsilon_0 \alpha \int_0^R \int_0^{2\pi} \int_0^\pi (4r e^{-\beta r} - \beta r^2 e^{-\beta r}) r^2 \sin \phi \, d\phi \, d\psi \, dr \\ &= 4\pi \epsilon_0 \alpha \int_0^R (4r^3 - \beta r^4) e^{-\beta r} dr \\ &= 4\pi \epsilon_0 \alpha [r^4 e^{-\beta r}]_0^R = 4\pi \epsilon_0 R^4 \alpha e^{-\beta R} \end{aligned}$$

$$\begin{aligned}
 \text{1b) } \frac{q_{\text{enc}}}{\epsilon_0} &= \int_{\partial V} E \, da = \int_{\partial V} ar e^{-\beta r} \cdot r \, da \\
 &= 4\pi R^2 \propto e^{-\beta R}
 \end{aligned}$$

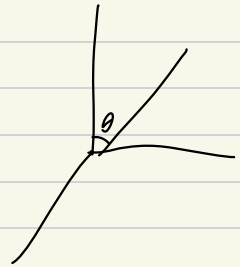
$$\begin{aligned}
 \text{3. (a) } E &= -\nabla V \\
 &= \begin{cases} -\frac{\alpha}{3\epsilon} (\cos\theta \hat{r} + \sin\theta \hat{\theta}) = -\frac{\alpha}{3\epsilon} \hat{z} \\ \frac{\alpha}{3\epsilon} \cdot \frac{R^3}{r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta}) \end{cases}
 \end{aligned}$$

$$\text{1b) } D = \epsilon E + P$$

$$P = \alpha \hat{z}$$

$$\sigma_b = P \cdot \hat{n} = \alpha \hat{z} \cdot \hat{n} = \alpha \cdot \frac{z}{R}$$

$$f_b = -\nabla \cdot P = 0$$



$$\text{(c) } \frac{\rho}{\epsilon} = \nabla \cdot E = 0$$

$$\text{4. (a) } J_b = \nabla \times M = \begin{vmatrix} \hat{r} & r\hat{\theta} & \hat{z} \\ \partial_r & \partial_\theta & \partial_z \\ 0 & \alpha r^3 & 0 \end{vmatrix} \frac{1}{r} = 3\alpha r \hat{z}$$

$$f_b = M \times \hat{r} = -\alpha R^3 \hat{z}$$

$$(b) \quad \nabla \times \vec{B} = \mu_0 \vec{J} = \mu_0 \nabla \times \vec{A}$$

$$\vec{B} \begin{cases} = \mu_0 M & z < R \\ = 0 & z > R \end{cases}$$

$$(c) \quad H = \frac{1}{\mu_0} B - M = 0$$

$$\text{f. (a) } \nabla \times E = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ * & 0 & 0 \end{vmatrix} = (0, \partial_z E, \partial_y E)$$

$$= (0, -k E_0 (\sin k(z-ct) + \sin k(z+ct)), 0)$$

$$\frac{\partial B}{\partial t} = k E_0 (\sin k(z-ct) + \sin k(z+ct))$$

(b)

$$\nabla \cdot B = 0 \quad \nabla \times B = \frac{1}{c^2} \frac{\partial E}{\partial t}$$

↓

$$\frac{\partial f}{\partial y} = 0$$

↓

$$\frac{\partial f}{\partial x} \cdot \frac{\partial f}{\partial y} = 0$$

$$f = c$$